CareConnect EHR System — REST API Specification

Project Code: EHRS01 | **Framework:** Spring Boot 3.5.7 (Java 24) RESTful Services
Base URL: `http://localhost:8080/api` | **Authentication:** Bearer JWT (`Authorization: Bearer <token>`)
Content-Type: `application/json` | **Error Model:** RFC 7807 Problem Details Standard

1. Authentication & Session Services

`POST` **`/auth/login`**

Authenticates user credentials and issues a signed role-scoped JWT token.

```
// Request Body
{
  "email": "doctor@apexcare.org",
  "password": "DoctorPass2026!",
  "role": "doctor"
}

// Response (200 OK)
{
  "token": "eyJhbGciOiJIUzM4NCJ9.eyJzdWIiOiIwMDAwMDAwMC0wMDAwLTAwMDAtMDAwMC0wMDAwMDAwMDAwMDIiLCJyb2xlIjoiRE9DV
  "user": {
    "id": "00000000-0000-0000-0000-000000000002",
    "name": "Dr. Rajesh Verma",
    "email": "rajesh.verma@apexcare.org",
    "role": "doctor"
  }
}
```

2. Patient Registration & Summary Views (Epic 1 & 6)

`POST` **`/patients/register-profile`**

Registers a new patient profile with demographics, insurance, and medical history.

```
// Request Body
{
  "firstName": "Vikram",
  "lastName": "Malhotra",
  "dob": "1975-08-23",
  "gender": "Male",
  "bloodGroup": "O+",
  "insuranceProvider": "HDFC ERGO Health",
  "insurancePolicyNumber": "HD-884102-M",
  "allergies": ["Sulfa Drugs"],
  "chronicConditions": ["Coronary Artery Disease"]
}
```

`GET` **`/views/patient-summary/{patientId}`**

Returns aggregated summary data matching HCL PatientSummary database view specifications.

```
// Response (200 OK)
{
  "patient": { "id": "pat-101", "name": "Ananya Sharma", "bloodGroup": "A+", "dob": "1988-04-12" },
  "activePrescriptionsCount": 2,
  "pendingLabOrdersCount": 1,
  "upcomingAppointmentsCount": 1
}
```

3. Clinical Encounter Documentation & Note Immutability (Epic 2)

`POST` **`/patients/{patientId}/clinical-notes`**

Creates a new draft or finalized clinical note with specialty template support.

```
// Request Body
{
  "doctorName": "Dr. Rajesh Verma",
  "templateType": "SOAP",
  "assessment": "Essential hypertension with good compliance; stable glycemic control.",
  "plan": "Continue current medications; schedule comprehensive lipid profile in 4 weeks.",
  "isFinalized": true
}
```

`PUT` **`/patients/{patientId}/clinical-notes/{noteId}`**

Note Immutability Enforcement: If `isFinalized == true`, returns HTTP 409 Conflict.

```
// Response when attempting to modify signed note (409 Conflict)
{
  "status": 409,
  "message": "Finalized clinical notes cannot be modified or overwritten.",
  "timestamp": "2026-10-30T18:22:00.785Z",
  "path": "/api/patients/pat-101/clinical-notes/cn-101"
}
```

4. Computerized Provider Order Entry (CPOE) (Epic 3)

`POST` **`/orders/lab`**

Places a laboratory diagnostic order with LOINC test coding and priority triage.

```
// Request Body
{
  "patientId": "pat-101",
  "orderingDoctorId": "doc-001",
  "testName": "Serum Potassium & Electrolytes",
  "testCode": "2823-3",
  "urgency": "STAT",
  "clinicalIndication": "Monitor ACE inhibitor therapy"
}
```

`POST` **`/orders/imaging`**

Orders radiological imaging studies with modality and body-site selection.

```
// Request Body
{
  "patientId": "pat-101",
  "modality": "MRI",
  "bodySite": "Brain with and without IV Contrast",
  "clinicalIndication": "Recurrent refractory cephalalgia"
}
```

5. Medication Management & Pharmacology Safety (Epic 4)

`POST` **`/prescriptions/check-interactions`**

Screens proposed prescriptions against active patient medications and documented drug allergies.

```
// Request Body
{
  "patientId": "pat-101",
  "proposedMedication": "Spironolactone",
  "currentMedications": ["Lisinopril"]
}

// Response (200 OK)
{
  "overallStatus": "CONTRAINDICATED",
  "interactions": [
    {
      "severity": "HIGH",
      "drugPair": "Lisinopril + Spironolactone",
      "clinicalRisk": "Risk of severe fatal hyperkalemia and cardiac arrhythmias."
    }
  ]
}
```

`POST` **`/patients/{patientId}/mar`**

Logs nurse-verified Medication Administration Records (MAR).

```
// Request Body
{
  "prescriptionId": "rx-201",
  "medicationName": "Atorvastatin 20mg",
  "doseAdministered": "20mg PO",
  "administeredBy": "Nurse Meera S.",
  "status": "GIVEN"
}
```

6. Diagnostic Integration & Standards Gateway (Epic 5)

`GET` **`/integrations/status`**

Returns the operational status of the CareConnect Diagnostic Interoperability Gateway.

`GET` **`/integrations/hl7/oru-r01/{orderId}`**

Returns standard HL7 v2.5.1 pipe-and-hat observation message.

```
MSH|^~\&|CARECONNECT_EHR|APEXCARE|LIS_GATEWAY|CENTRAL_LAB|202610302345||ORU^R01|MSG-9921|P|2.5.1
PID|1||pat-101^^^MRN||Sharma^Ananya||19880412|F
OBR|1|ORD-101||2823-3^Potassium Serum^LN|||202610302330
OBX|1|NM|2823-3^Potassium^LN||6.4|mmol/L|3.5-5.0|C|||F
```

`GET` **`/integrations/fhir/diagnostic-report/{orderId}`**

Returns HL7 FHIR Release 4 DiagnosticReport and Observation JSON bundle.